

Production Trends and Asset Concentration Analysis in Midland County Using Texas Railroad Commission Public Data

SPE-Style Independent Technical Report

Joseph Luongo
The University of Texas at Austin
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Study Area: Midland County, Texas
Study Period: January 2020 - March 2026
Data Source: Texas Railroad Commission Production Data Query exports

Table 1 - Executive Summary Metrics

Metric	Result
Total Oil	1,379,471,252 BBL
Total Gas	4,860,365,156 MCF
Average GOR	3.52 MCF/BBL
2020-2025 Oil Growth	26.1%
2020-2025 Gas Growth	78.7%
Top Field	SPRABERRY (TREND AREA)
Top Operator	PIONEER NATURAL RES. USA, INC.
Top 10% Lease Oil Share	52.0%

Abstract

Midland County, Texas is one of the most active production areas in the Midland Basin portion of the Permian Basin. This study evaluates public production data obtained from Texas Railroad Commission Production Data Query exports for the period January 2020 through March 2026. The analysis uses county-level monthly totals, operator-level summaries, field-level summaries, and lease-level production records to identify production trends, operator performance, field contribution, gas-oil ratio behavior, and lease-level concentration. Midland County produced 1,379,471,252 BBL of oil and 4,860,365,156 MCF of gas over the study period. Annual oil production increased from 191,675,933 BBL in 2020 to 241,692,085 BBL in 2025, representing 26.1% growth. Gas production increased from 549,802,603 MCF in 2020 to 982,604,254 MCF in 2025, representing 78.7% growth. Pioneer Natural Resources USA, Inc. was the largest operator by oil production, and Spraberry (Trend Area) was the largest producing field. Lease-level analysis indicates that the highest-producing 10% of producing leases generated approximately 52.0% of county oil production. These results demonstrate that Midland County remained a growing oil province during the study period while becoming increasingly gas-rich on an aggregate basis.

1. Introduction

The Permian Basin remains one of the most important oil and gas producing provinces in North America. Midland County is located in the Midland Basin subregion of the Permian and is associated with prolific unconventional resource development, especially in Spraberry and Wolfcamp-related intervals. Because the county is heavily developed and has a large number of producing leases and operators, it is a useful study area for understanding regional production behavior from public data.

The purpose of this report is to transform public regulatory production data into a structured technical analysis. Rather than relying on proprietary databases, this project uses Texas Railroad Commission records that can be accessed by students, engineers, analysts, and the public. The main engineering value of this type of work is not simply plotting production volumes, but interpreting what production trends suggest about development intensity, gas-oil ratio changes, field concentration, and the distribution of output across operators and leases.

The study focuses on Midland County only. This narrower scope makes the project manageable while still providing enough data for meaningful conclusions. The same workflow can later be expanded to Martin, Reeves, Loving, Howard, Upton, or Glasscock Counties to create a broader Permian Basin benchmarking study.

2. Objectives

The objectives of this study were to evaluate county-wide production trends, compare operator and field contributions, and quantify production concentration at the lease level. The analysis was designed to answer five practical questions: (1) how did oil production change from 2020 to 2025, (2) how did gas production change over the same period, (3) which operators contributed the most oil production, (4) which fields dominated county production, and (5) how concentrated was production among the highest-producing leases?

A secondary objective was to create a professional analysis package that could be used in a resume, LinkedIn post, GitHub portfolio, or interview discussion. For that reason, the report emphasizes reproducible methods, clear figures, and conclusions tied directly to measured data.

3. Data Source and Scope

The data used in this report were downloaded from Texas Railroad Commission Production Data Query exports for Midland County. The imported files included a Monthly Totals Report, Operator Report, Field Report, Lease Report, and District Report. The Texas Railroad Commission states that Production Data Query provides oil and gas production information by lease, field, operator, district, or county from January 1993 to present. For this project, the selected data period was January 2020 through March 2026.

The workbook used for this report contained 75 monthly records, 7 annual summary rows, 20 ranked operators, 20 ranked fields, and 20 ranked leases. The complete field dataset contained 94 field-level records. The source workbook also notes that the lease-level dataset contained 4,240 nonzero-oil leases and that the original lease-level export contained 5,675 lease records. These data were aggregated by month, operator, field, and lease to produce the results presented in this report.

The analysis is based on reported production volumes and does not include completion design parameters, lateral length, proppant loading, formation benches, working interest, commodity price, or lease operating expense. Therefore, the report should be interpreted as a production-data study, not a full reservoir characterization or economic evaluation.

4. Methodology

The workflow consisted of data organization, cleaning, aggregation, graphing, and interpretation. The raw Texas Railroad Commission exports were organized into separate categories by reporting view. Monthly totals were used for county-wide time-series analysis. Operator totals were used to rank companies by cumulative oil production. Field totals were used to rank producing fields. Lease totals were used to evaluate how much of the county's production came from the highest-producing assets.

Oil volumes were measured in barrels (BBL). Gas volumes were measured in thousand cubic feet (MCF). Gas-oil ratio (GOR) was calculated as total gas production divided by total oil production, with units of MCF/BBL. The GOR trend was evaluated over time because it helps indicate whether aggregate county production is becoming more gas-rich relative to oil output.

Annual growth rates were calculated by comparing full-year 2020 production with full-year 2025 production. Year 2026 was not used in the main growth comparison because the dataset only contains January through March 2026. Lease concentration was evaluated by ranking leases from highest to lowest oil production and calculating the share of total county oil production attributable to the highest-producing leases.

5. Results

5.1 County-Level Production Trends

County-wide oil production increased from 191,675,933 BBL in 2020 to 241,692,085 BBL in 2025. This represents 26.1% growth over the full-year comparison period. Gas production increased from 549,802,603 MCF in 2020 to 982,604,254 MCF in 2025, a 78.7% increase. Because gas grew much faster than oil, the aggregate GOR increased from 2.87 MCF/BBL in 2020 to 4.07 MCF/BBL in 2025.

Table 2 - Midland County Annual Production Summary

Year	Months in Dataset	Oil (BBL)	Gas Total (MCF)	Condensate (BBL)	GOR (MCF/BBL)
2,020.00	12.00	191,675,933.00	549,802,603.00	68,507.00	2.87
2,021.00	12.00	199,143,122.00	630,968,700.00	69,577.00	3.17
2,022.00	12.00	221,678,081.00	720,013,945.00	63,259.00	3.25
2,023.00	12.00	234,518,111.00	829,080,596.00	59,219.00	3.54
2,024.00	12.00	234,762,267.00	915,274,251.00	50,801.00	3.90
2,025.00	12.00	241,692,085.00	982,604,254.00	764,880.00	4.07
2,026.00	3.00	56,001,653.00	232,620,807.00	188,718.00	4.15

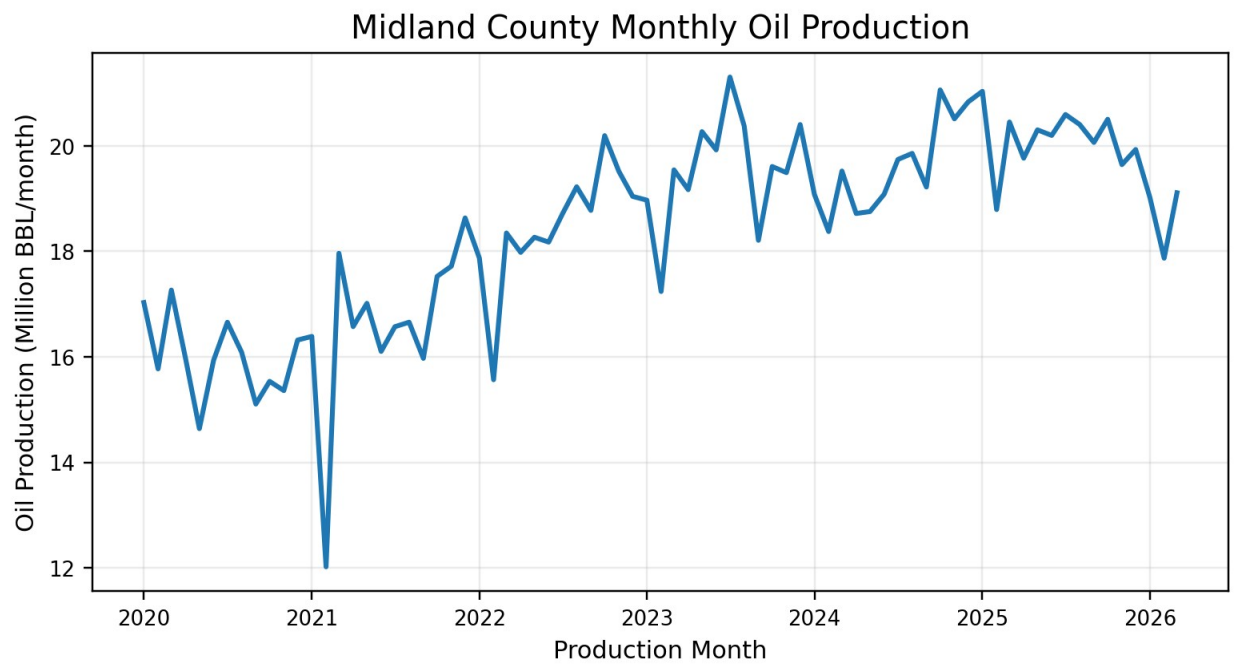


Figure 1 - Monthly oil production in Midland County from January 2020 through March 2026. Full-year annual oil production increased 26.1% from 2020 to 2025.

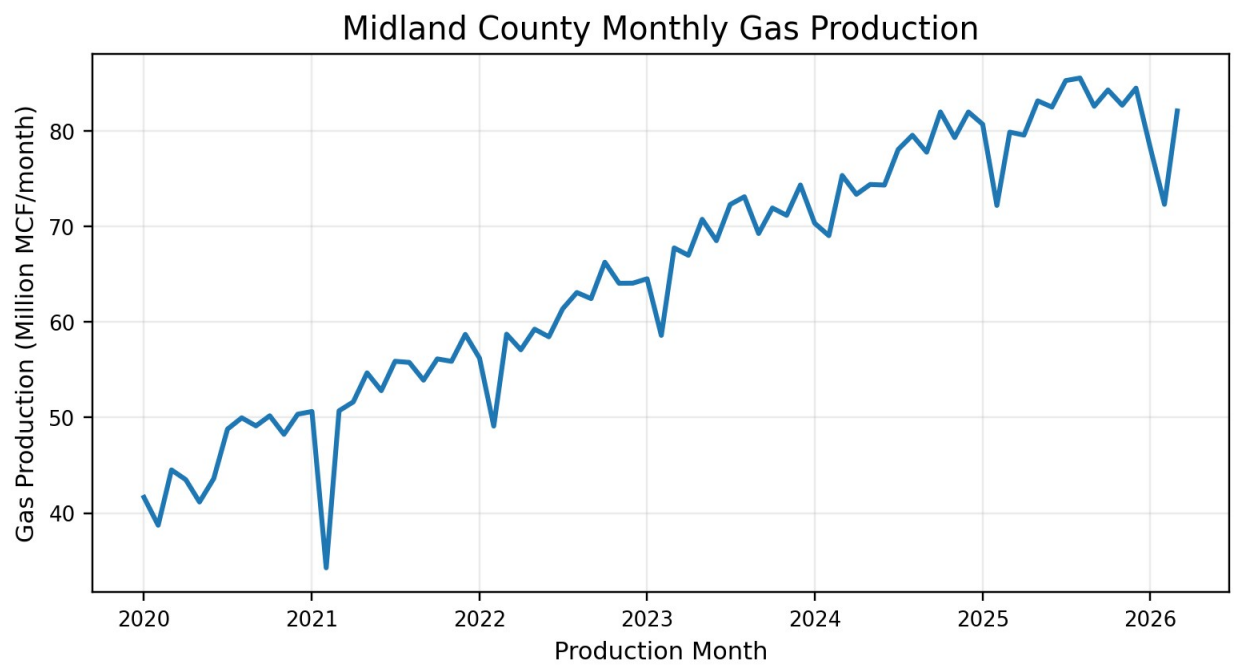


Figure 2 - Monthly gas production in Midland County from January 2020 through March 2026. Full-year annual gas production increased 78.7% from 2020 to 2025.

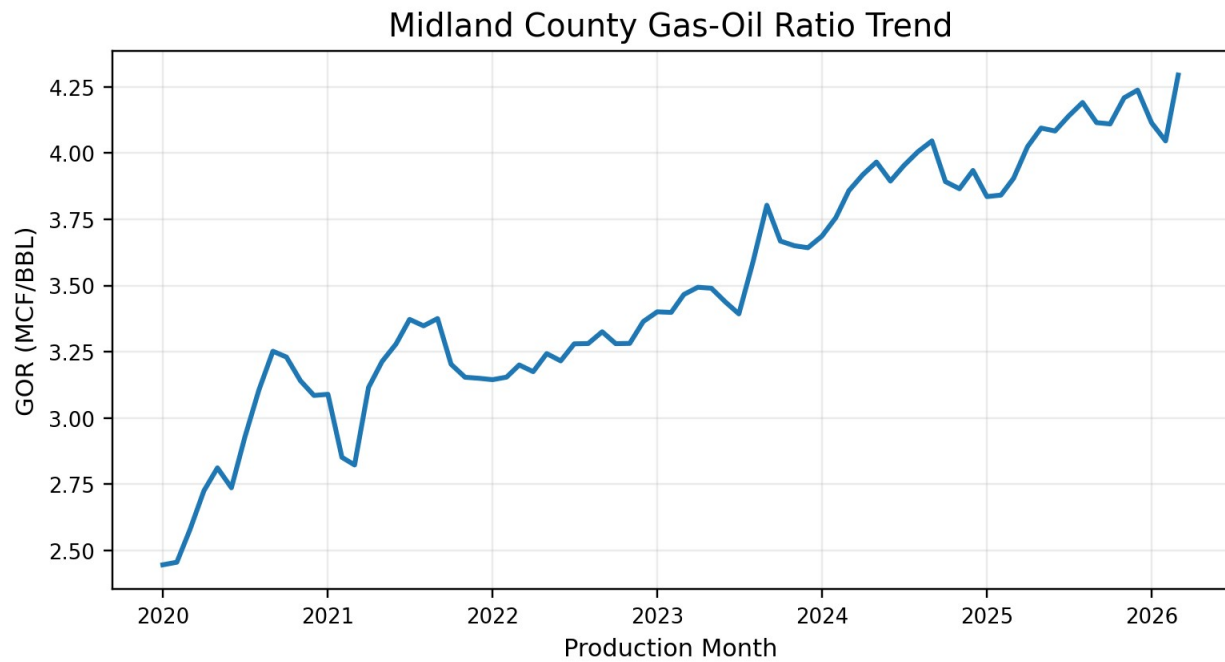


Figure 3 - Monthly gas-oil ratio trend. The increase in GOR indicates that aggregate Midland County production became more gas-rich during the study period.

5.2 Operator-Level Analysis

Operator-level results show that production is highly concentrated among a relatively small group of companies. Pioneer Natural Resources USA, Inc. ranked first with 461,756,167 BBL of oil, representing 33.5% of county oil production in the analysis workbook. Endeavor Energy Resources L.P. ranked second, and XTO Energy Inc. ranked third. The top five operators together represented 71.3% of county oil production.

Table 3 - Top 10 Operators by Oil Production

Rank	Operator Name	Oil (BBL)	Gas Total (MCF)	GOR (MCF/BBL)	Share of County Oil
1	PIONEER NATURAL RES. USA, INC.	461,756,167	1,573,697,233	3.41	33.5%
2	ENDEAVOR ENERGY RESOURCES L.P.	151,813,457	437,944,654	2.88	11.0%
3	XTO ENERGY INC.	144,255,030	698,852,985	4.84	10.5%
4	CHEVRON U. S. A. INC.	114,741,932	359,035,478	3.13	8.3%
5	DIAMONDBACK E&P LLC	111,512,115	469,461,067	4.21	8.1%
6	COG OPERATING LLC	80,378,990	270,902,918	3.37	5.8%
7	APACHE CORPORATION	59,634,834	186,858,394	3.13	4.3%
8	OVINTIV USA INC.	45,012,707	235,748,375	5.24	3.3%
9	CROWNQUEST OPERATING, LLC	36,465,637	139,906,375	3.84	2.6%
10	PERMIAN DEEP ROCK OIL CO., LLC	32,774,650	70,023,183	2.14	2.4%

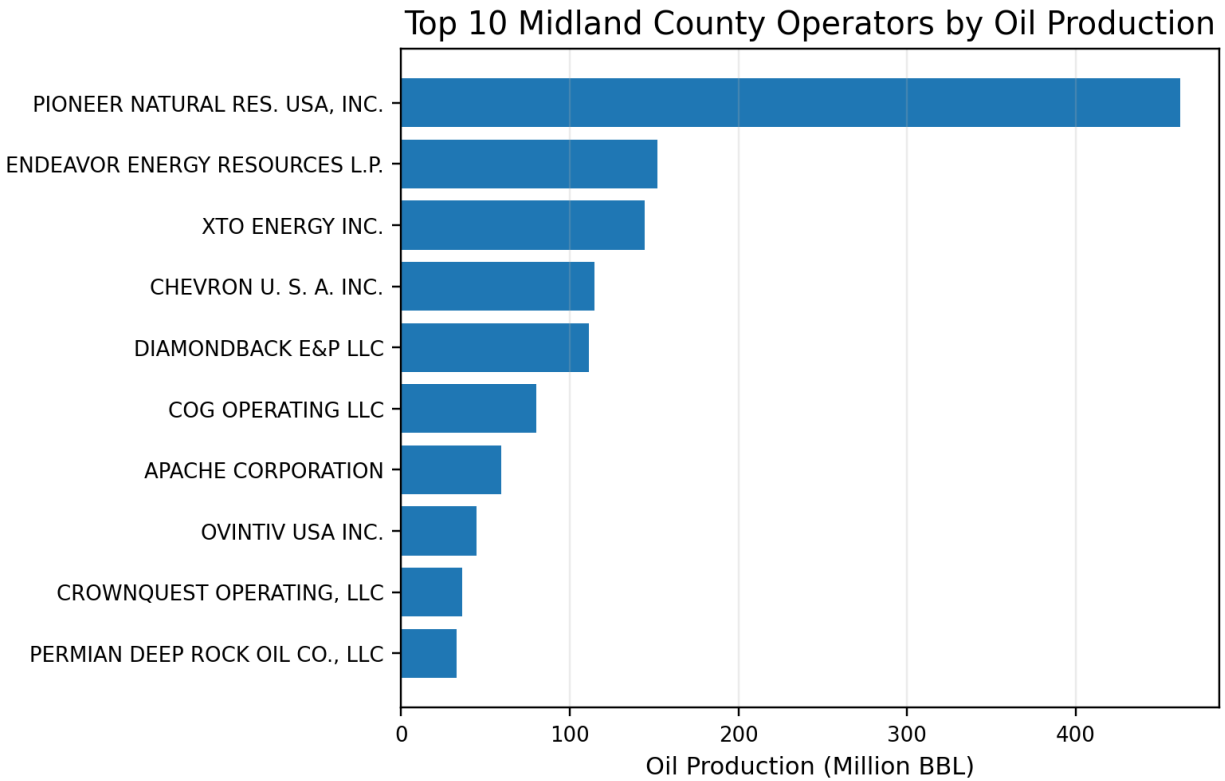


Figure 4 - Top 10 Midland County operators by cumulative oil production in the study dataset.

5.3 Field-Level Analysis

Field-level results show an even stronger concentration pattern. Spraberry (Trend Area) was the dominant field, producing 1,119,241,146 BBL of oil and accounting for 81.1% of county oil in the analysis workbook. The second-largest field was Spraberry (Trend Area) R 40 EXC, followed by Parks (Consolidated). Together, the top three fields accounted for 98.4% of county oil production.

Table 4 - Top 10 Fields by Oil Production

Rank	Field Name	Oil (BBL)	Gas Total (MCF)	GOR (MCF/BBL)	Share of County Oil
1	SPRABERRY (TREND AREA)	1,119,241,146	3,628,915,534	3.24	81.1%
2	SPRABERRY (TREND AREA) R 40 EXC	129,676,142	787,274,092	6.07	9.4%
3	PARKS (CONSOLIDATED)	108,221,412	294,759,617	2.72	7.8%
4	PEGASUS (CONSOLIDATED)	14,920,092	87,644,426	5.87	1.1%
5	EMMA (BARNETT SHALE)	1,562,968	11,774,484	7.53	0.1%
6	DORA ROBERTS (CONSOLIDATED)	1,524,912	9,569,679	6.28	0.1%
7	SPRABERRY (TREND AREA)	1,425,332	3,369,559	2.36	0.1%
8	SPRABERRY (TREND AREA) R 40 EXC	703,093	12,792,270	18.19	0.1%
9	GERMANIA (GRAYBURG)	275,137	33,697	0.12	0.0%
10	PEART (BARNETT)	267,373	1,986,730	7.43	0.0%

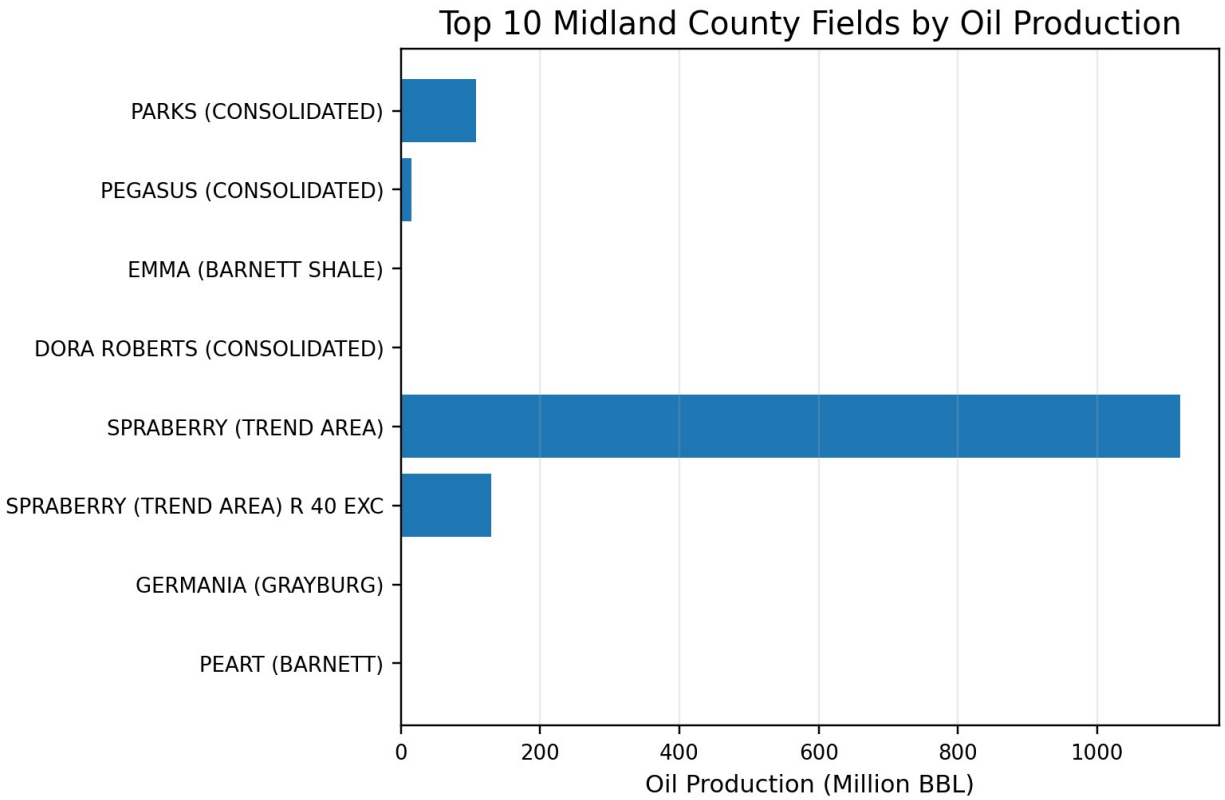


Figure 5 - Top 10 Midland County fields by cumulative oil production in the study dataset.

5.4 Lease-Level Concentration Analysis

Lease-level production was not evenly distributed. The source workbook identified 4,240 nonzero-oil leases, and the highest-producing 10% of producing leases accounted for approximately 52.0% of county oil production. The top 20 leases alone accounted for approximately 10.5% of county oil. This indicates that a relatively small number of assets are responsible for a disproportionate amount of production.

Table 5 - Top 10 Leases by Oil Production

Rank	Lease Name	Oil (BBL)	Gas Total (MCF)	GOR (MCF/BBL)	Share of County Oil
1	TEXAS TEN Y PU	18,773,680	79,591,550	4.24	1.36%
2	SOA SCHARBAUER SE	13,745,848	41,746,441	3.04	1.00%
3	SCHARBAUER, C.	11,007,423	29,688,179	2.70	0.80%
4	NOBLES	8,536,846	17,812,571	2.09	0.62%
5	SOUTH CURTIS RANCH	8,223,656	15,634,889	1.90	0.60%
6	PEGASUS FIELD UNIT 3	7,595,628	51,123,303	6.73	0.55%
7	BRD ASTRODOME UNIT	7,453,582	27,339,206	3.67	0.54%
8	GBG REEDY 13/25 UNIT	7,135,703	25,240,577	3.54	0.52%
9	HUTT, DONALD L'C'	6,668,492	35,217,798	5.28	0.48%
10	GRIDIRON	5,909,076	20,968,411	3.55	0.43%

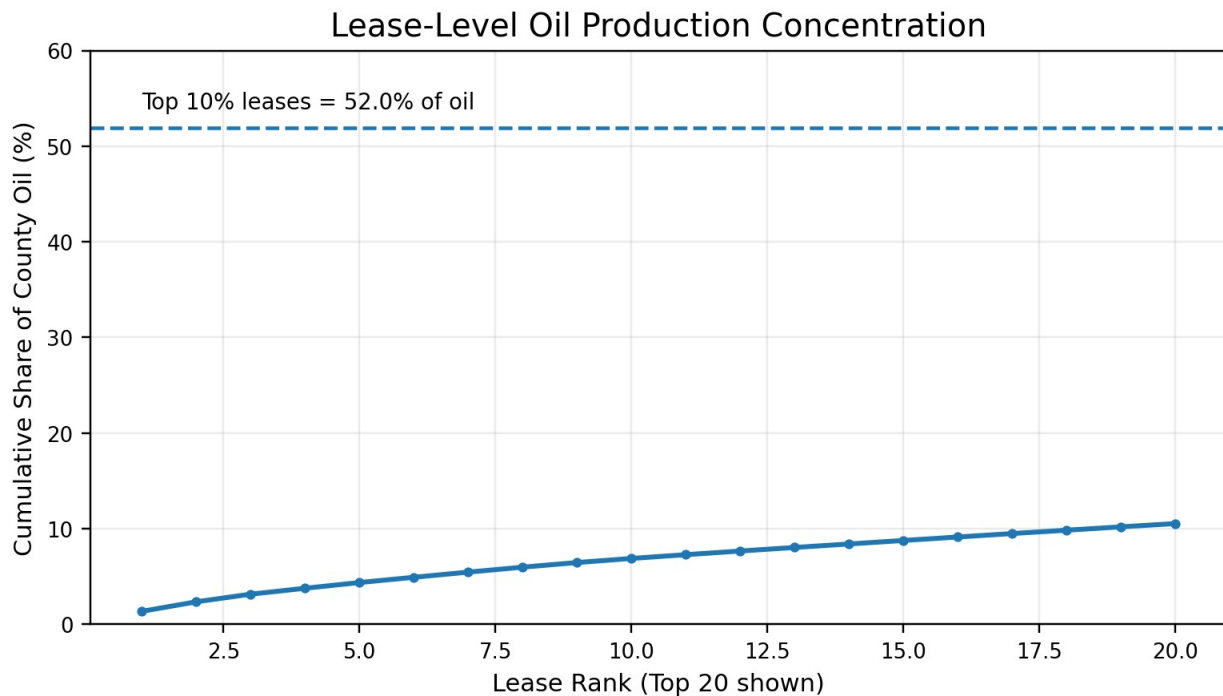


Figure 6 - Lease-level concentration view. The top 20 leases are shown, with the reported top 10% lease share marked for context.

6. Discussion

The most important result is that Midland County continued to grow in both oil and gas production between 2020 and 2025. The oil growth rate of 26.1% is substantial given that Midland County is already a mature producing area with a long development history. This suggests that modern unconventional development, recompletions, infill activity, and continued capital deployment kept county production moving upward during the study period.

The gas trend is even more notable. Gas production increased 78.7% from 2020 to 2025, far outpacing oil growth. As a result, aggregate GOR increased over time. This does not prove a single reservoir mechanism by itself, but it does suggest that the county's reported production stream became more gas-rich on an aggregate basis. Potential explanations include increasing production from higher-GOR wells, changes in completion strategy, development of areas with different fluid properties, or the natural evolution of unconventional wells as they mature. A well-level decline study would be required to separate these effects.

The operator and field analyses show that Midland County production is concentrated. Pioneer Natural Resources USA, Inc. was the largest operator by oil production, while Spraberry (Trend Area) was by far the dominant field. This field-level concentration is consistent with the regional importance of the Spraberry trend in the Midland Basin. From an engineering perspective, this means that county-level production performance is heavily influenced by a small set of major producing intervals and operators.

The lease concentration result is also important. The highest-producing 10% of leases generated approximately 52.0% of oil production. This is a practical reminder that not all assets contribute equally. Differences in geology, development spacing, completion quality, reservoir pressure, operating practices, and well count can create large production differences between leases. For a company evaluating acquisition targets or development priorities, identifying high-contribution leases would be one of the first steps in understanding asset value.

7. Limitations

This report is based on aggregated Texas Railroad Commission query exports and therefore has several limitations. First, the data provide production volumes but do not include all engineering variables required for a complete reservoir study. Completion dates, lateral lengths, proppant intensity, frac fluid volumes, choke management, formation bench, pressure history, and well spacing were not included in the workbook used for this report.

Second, the analysis does not perform normalized well-level comparisons. Without well counts, lateral lengths, and start dates, a lease with high cumulative production may appear strong simply because it contains more wells or older producing wells. Future work should normalize production by well count, first production date, and lateral length if those data can be joined from completion reports or well records.

Third, the 2026 data are partial because the dataset only includes January through March 2026. For that reason, 2026 was included in charts but excluded from the main annual growth comparison. Finally, the analysis is limited to Midland County and should not be generalized to the entire Permian Basin without comparing additional counties.

8. Future Work

This project can be expanded in several ways. The most direct improvement would be to repeat the workflow for Martin, Reeves, Loving, Howard, Upton, and Glasscock Counties. That expansion would turn the project into a broader Permian Basin benchmarking study and would make it possible to compare Midland Basin and Delaware Basin production behavior.

A second improvement would be to join lease-level production data with well records and completion information. That would allow decline curve analysis, well vintage analysis, and normalized production comparisons. For example, wells could be grouped by first production year and compared on 6-month, 12-month, and 24-month cumulative production.

A third improvement would be to build a simple forecasting or machine-learning model. Possible target variables could include first-year cumulative oil, first-year gas production, or future GOR. Predictor variables could include county, operator, field, completion year, lateral length, and historical production. Even a basic regression model would add a useful data science component to the project.

9. Conclusions

This study used Texas Railroad Commission public production data to evaluate Midland County production from January 2020 through March 2026. The main findings are summarized below.

1. Midland County produced 1,379,471,252 BBL of oil and 4,860,365,156 MCF of gas during the study period.
2. Oil production increased 26.1% from 2020 to 2025.
3. Gas production increased 78.7% from 2020 to 2025, causing aggregate GOR to rise.
4. Pioneer Natural Resources USA, Inc. was the highest-producing operator by oil production.
5. Spraberry (Trend Area) was the dominant field, representing 81.1% of county oil in the workbook.
6. The highest-producing 10% of producing leases accounted for approximately 52.0% of county oil production.

Overall, the analysis shows that publicly available RRC data can be used to produce meaningful production insights. The results also demonstrate that Midland County remained a strong producing region while showing increasing gas intensity during the 2020-2025 period.

References

Railroad Commission of Texas. Production Data Query System. <https://webapps2.rrc.texas.gov/EWA/ewaPdqMain.do>

Railroad Commission of Texas. Production Data Query System FAQs. <https://www.rrc.texas.gov/about-us/faqs/oil-gas-faq/production-data-query-system-faqs/>

Railroad Commission of Texas. Oil & Gas Production Data.
<https://www.rrc.texas.gov/oil-and-gas/research-and-statistics/production-data/>

Society of Petroleum Engineers. Author Resources and Paper Preparation Guidance.
<https://www.spe.org/en/authors/resources/prepare-paper/>

Appendix A - Data Files Used

The analysis was based on the following user-downloaded Texas Railroad Commission CSV exports:

- ProductionDataQuery_monthlyTotals_ReportCsv.csv
- ProductionDataQuery_Operator_ReportCsv.csv
- ProductionDataQuery_Field_ReportCsv.csv
- ProductionDataQuery_Lease_ReportCsv.csv
- ProductionDataQuery_District_ReportCsv.csv